Individual Homework Assignment

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Assignment theme: Pizza ratings and sales

Data files. Use exactly these files from your `student_files` folder:

- Task 1: `data_task1_variant2.csv`
- Task 2: `data_task2_variant2.csv`
- Task 3: `data_task3_variant2.csv`

Task 1 – One-way comparison of group means (5 points)

The file `data_task1_variant2.csv` contains data on different pizza types and their customer ratings. The file has two columns:

- `Score`: the customer rating of a pizza on a 0-10 scale;
- `Group`: the pizza type. Possible groups: `Margherita`, `Pepperoni`, `Hawaiian`, `Quattro Formaggi`, `Vegetarian`.

1. **Hypothesis test.** Test, at significance level $\alpha = 0.05$, whether the mean values of `Score` differ significantly across the groups.
2. **Post-hoc comparisons.** If the test is significant, perform pairwise post-hoc comparisons and comment carefully on which groups differ.

Task 2 – Multiple linear regression (20 points)

The file `data_task2_variant2.csv` contains data on pizzas. Fit a multiple linear regression model where Y is the weekly number of pizzas sold, X_1 is the price of the pizza, measured in Hungarian forints, and X_2 is the customer rating of the pizza.

1. Estimation (7 points)

1. **Coefficient estimates (3 points).** Compute the point estimates and standardized coefficients. Interpret them and write down the fitted model.
2. **Prediction (1 point).** Predict the response for a pizza with price $X_1 = 1500$ HUF and customer rating $X_2 = 8.5$.
3. **Confidence intervals (2 points).** Compute 95% confidence intervals for the coefficients and a 95% confidence interval for the expected response at the prediction point. Interpret them.
4. **Prediction interval (1 point).** Compute a 95% prediction interval for an individual observation at the same prediction point.

2. Goodness of fit (3 points)

Compute and interpret R^2 and adjusted R^2. Explain why adjusted R^2 is useful in multiple regression.

3. Model diagnostics (10 points)

1. Test the overall significance of the model at the 5% level.
2. Test the significance of the individual coefficients at the 5% level.
3. Compute VIF values and discuss multicollinearity.
4. Check the residual assumptions: zero mean, normality, independence, and homoscedasticity. Estimate the error variance.

Task 3 – Introductory time-series modelling (15 points)

The file `data_task3_variant2.csv` contains a time series describing monthly sales data from a pizza restaurant. It has columns `time` and `value`.

1. **Deterministic model (5 points).** Fit a deterministic trend model, compute the coefficients, check residual assumptions, and forecast the next few months.
2. **Exponential smoothing (5 points).** Apply exponential smoothing, evaluate the fit, and forecast the next few months.
3. **Box–Jenkins model (5 points).** Fit an ARIMA or SARIMA model, examine parameters and residuals, and forecast the next few months.